

SARDAR PATEL UNIVERSITY
B.SC. (CA & IT) SIXTH SEMESTER (CBCS) EXAMINATION 2021
PS06CIIT25 - ARTIFICIAL INTELLIGENCE
24th JULY 2021

Time: 02:00 PM to 04:00 PM

Marks: 70

Q.1 Multiple Choice Questions

[10]

1. Medical diagnosis is a _____.
a) Mundane Task
c) Expert Task
b) Formal Task
d) None
2. The collection of triggered rules is known as a _____.
a) inference engine
c) conflict resolution
b) conflict set
d) none
3. A _____ is a string of symbols that is generated by a formal language.
a) propositional logic
c) well-formed formula
b) first-order logic
d) none
4. _____ knowledge represents how to reach a solution in a given situation.
a) factual
c) expert
b) procedural
d) none
5. In _____ AI language, facts and procedures are developed and presented in the form of list.
a) Java
c) LISP
b) PROLOG
d) none
6. _____ converts the fuzzy value into a Crisp Value.
a) Fuzzification
c) normalization
b) Defuzzification
d) inference
7. The learning rule of perceptron is also known as _____.
a) Delta rule
c) Gama rule
b) Beta rule
d) Alpha rule
8. _____ is one of the key operators of genetic algorithm it is the process of selecting individual from a population for reproduction based on their fitness values.
a) Selection
c) Mutation
b) Crossover
d) Inversion
9. _____ is a collection of such genes representing possible solutions in domain space.
a) Genetic Cycle
c) Genotype
b) Genetic algorithm
d) Mutation
10. DNA consist of string built by _____ responsible for some property of human to which it belongs.
a) genes
c) Stored program concept
b) number
d) None

Q.2 Fill in the Blanks and True or False.

[08]

1. The test in which the computer mimics a human and tries to fool other humans is called _____.
2. The _____ process emphasizes capturing experts' thoughts and experiences.
3. ANN stands for _____.
4. The principle of Genetic Algorithm is based on _____.
5. KBS is an example of fifth-generation computer technology [True/False]
6. Self-updates is known as automatic learning or machine learning. [True/False]
7. A perceptron may contain more than one layer to solve complex real-world application. [True/False]
8. The degree of weakness is referred to the strength of the genes in an individual. [True/False]

Q.3 Answer the following questions. (Any ten)

[20]

1. Explain Chinese Room Experiment.
2. Draw diagram of Data pyramid according to management perspective.
3. State problems in hill climbing algorithm.
4. Explain the difficulties in KBS development.
5. Explain the following terms:
 1. Protocol analysis
 2. Knowledge representation
6. Write the steps of knowledge acquisition in KBS.
7. What is Fuzzy Inference?
8. Explain Linguistic Variables.
9. What is Fuzzy Rule Based System (FRBS)? List two models available for FRBS.
10. Define Chromosomes and Genotype.
11. What is Genetic Algorithms? Write the Genetic Algorithm.
12. What is Mutation?

Q.4 Answer the following questions. (Any Four)

[32]

1. Explain data pyramid in detail.
2. List main types of KBS and explain any three in detail.
3. Explain in detail the knowledge-based system development model along with diagram.
4. Write a note on knowledge acquisition.
5. Explain Membership Functions in detail.
6. Explain Operations on Fuzzy Sets.
7. Explain Travelling Sales Person problem in detail.
8. Explain Crossover in detail.

—X—

(2)